

Assignment 02 (R) ST2137-2520

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Dataset

The dataset `FEV.DAT` contains pulmonary function measures on 654 children ages 3 to 19 seen in East Boston, Massachusetts as part of a Childhood Respiratory Disease study. The dataset contains measurements on age, sex, height (in inches), smoking status, and FEV. FEV stands for Forced Expiration Volume. It is the volume of air, in litres, expelled in 1 second.

These are the columns in the dataset:

- FEV: forced expiratory volume in liters (Y)
- Age: age in years (X_1)
- Ht: height measured in inches (X_2)
- Gender: gender (0=female, 1=male) (X_3)
- Smoke: exposure to smoking (0=no, 1=yes). (X_4)

1. Read the data into R as a dataframe `fev`, and convert the columns `Gender` and `Smoke` to factors.
2. Fit the following linear model:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 I(X_3 = 1) + \beta_4 I(X_4 = 1) + e$$

Extract the following values:

- `adj_r2`: The adjusted R^2 for the model

- `sigma2`: The estimate of σ^2 from the data.
- `fstat`: The F -statistic for the hypothesis test

$$H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = 0 \quad (1)$$

$$H_1 : \text{Not all } \beta_i = 0 \quad (2)$$

3. The following panel of scatterplots, in Figure 1 was created with the following command. Extract the points corresponding to the lower left-panel (Female Non-Smokers) and use it to display the regression line corresponding to the 10-th, 50-th and 90-th percentiles of Age, just like in Figure 2.

```
xyplot(FEV ~ Ht | Gender * Smoke, data=fev)
```

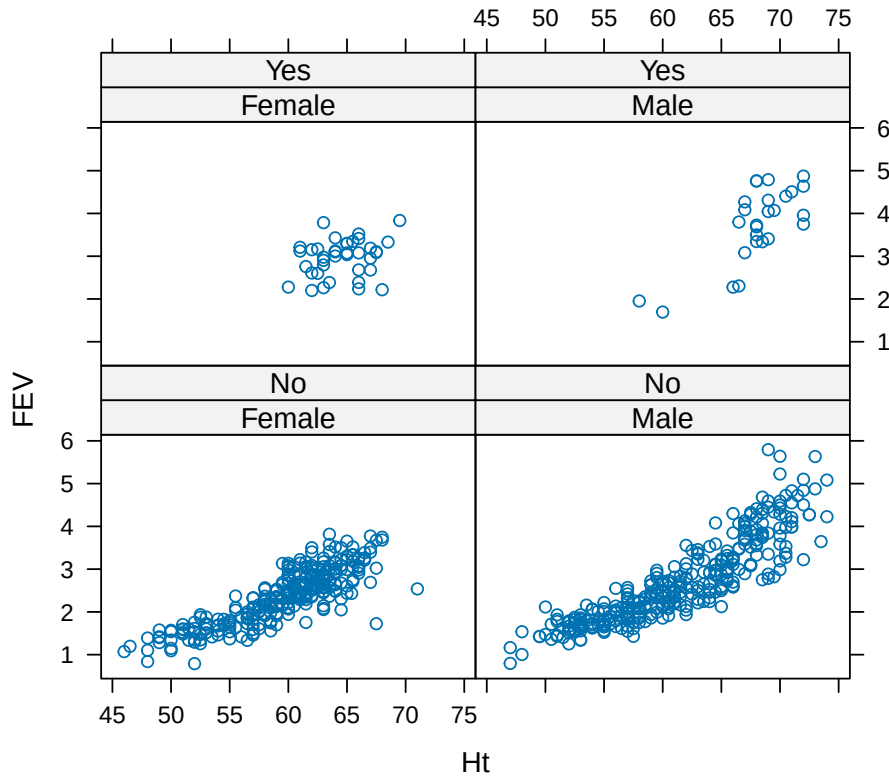


Figure 1: Scatter plot of FEV against Height

4. In tutorial 8, we computed a robust version of skewness. From the same paper, Kim and White (2004), here is a robust measure of kurtosis:

$$k_2 = \frac{(E_7 - E_5) + (E_3 - E_1)}{E_6 - E_2} - 1.23$$

FEV for Female Non-smokers, along with fitted regression line for Age quantiles

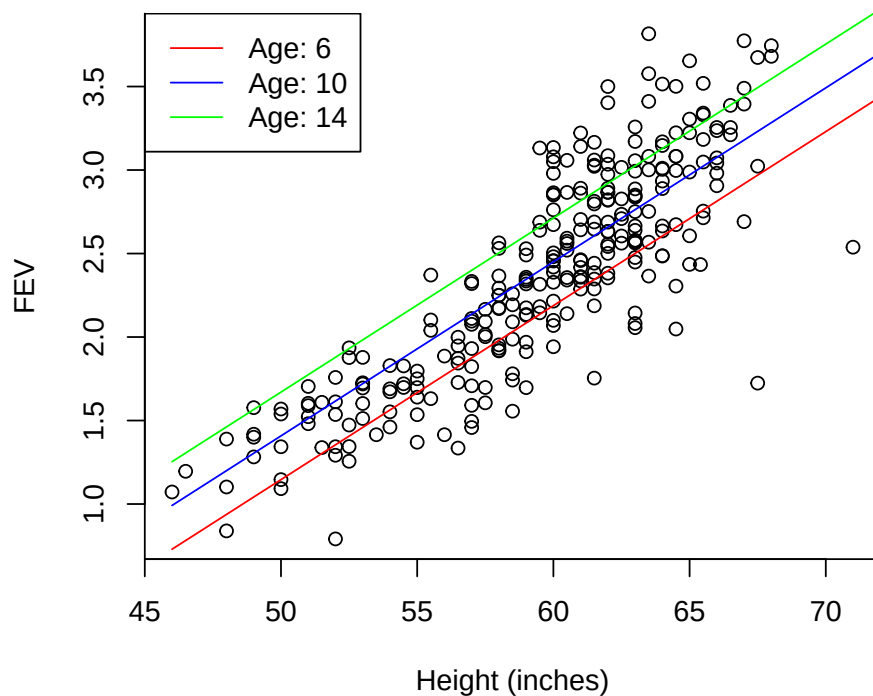


Figure 2: Fitted regression lines

where E_i is the i -th octile. The subtraction of 1.23 ensures that k_2 is relative to a Normal distribution: if $k_2 > 0$, it means that tails are fatter than those of a Normal. Write an R function named `robust4`. It should take in a numeric vector, and return a vector of length 4, corresponding to:

- trimmed mean, with $\gamma = 0.1$.
- MAD, with `constant=1.4826`
- Robust measure of skewness (h_2)
- Robust measure of kurtosis (k_2)

Use `aggregate` to apply this function to the standardised residuals from the above model (obtained using `rstandard`), grouped by all Gender x Smoke cross-combinations.

Requirements

Your code should create the following objects:

1. `fev` dataframe, with columns `FEV`, `Age`, `Ht`, `Gender`, `Smoke`.
 - `Gender` and `Smoke` must be factors.
2. `adj_r2`, `sigma2` and `fstat` scalars.
3. A scatterplot with the abovementioned fitted regression lines.
4. A function named `robust4` that returns a vector of length 4.

References

Kim, Tae-Hwan, and Halbert White. 2004. "On More Robust Estimation of Skewness and Kurtosis." *Finance Research Letters* 1 (1): 56–73.